

Algorithm for simultaneous Control of Stepper motors.

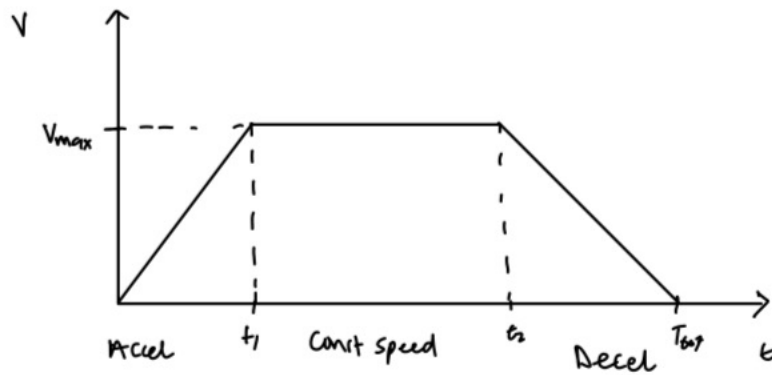
1. Calculate the difference between the final joint position and the initial position.
2. Find the largest joint delta and divide all the joints by it. (Gradient)
3. When you move the joint with the largest joint delta by 1, you move the other joints by their respective gradients

Implementing acceleration and deceleration

To reduce number of inputs to only T_{tot} .

* It will accelerate and decelerate for 50% of the time

* V_{max} is automatically calculated.



$$t_1 = 0.25 \times T_{tot} \quad t_2 = 0.75 \times T_{tot}$$

$$\Delta J_{max} = \left(\frac{1}{2} V_{max} \times t_1 \right) \times 2 + (t_2 - t_1) V_{max}$$

$$\Delta J_{max} = V_{max} t_1 + t_2 V_{max} - t_1 V_{max}$$

$$\Delta J_{max} = t_2 V_{max}$$

$$V_{max} = \frac{S}{t_2}$$

$$\Delta J_{max} = \text{Joint with max delta}$$

$$a = \frac{V_{max}}{t_1}$$

⚠ We can't control speed of stepper motor directly and so the time delay between each step should be controlled

When accelerating

$$S = \frac{1}{2} a t^2$$

$$t = \sqrt{\frac{2S}{a}}$$

$$\text{time delay} = (\sqrt{S+1} - \sqrt{S}) \sqrt{\frac{2S}{a}}$$

When decelerating

$$S = ut + \frac{1}{2} at^2 \quad D = \text{Distance travelled}$$

$$S = D + V_{max} t + \frac{1}{2} (-a) t^2$$

$$S = -\frac{a}{2} t^2 + V_{max} t + D$$

Completing Squares

$$S = -\frac{a}{2} \left[t^2 - \frac{2V_{max}}{a} t - \frac{2D}{a} \right]$$

$$S = -\frac{a}{2} \left[\left(t - \frac{V_{max}}{a} \right)^2 - \frac{2D}{a} - \left(-\frac{V_{max}}{a} \right)^2 \right]$$

$$S = -\frac{a}{2} \left(t - \frac{V_{max}}{a} \right)^2 + D + \frac{V_{max}^2}{2a}$$

$$S - D - \frac{V_{max}^2}{2a} = -\frac{a}{2} \left(t - \frac{V_{max}}{a} \right)^2$$

$$-\frac{2}{a} \left(S - D - \frac{V_{max}^2}{2a} \right) = \left(t - \frac{V_{max}}{a} \right)^2$$

$$\sqrt{-\frac{2S}{a} + \frac{2D}{a} + \frac{V_{max}^2}{a^2}} = t - \frac{V_{max}}{a}$$

$$t = \sqrt{-\frac{2S}{a} + \frac{2D}{a} + \frac{V_{max}^2}{a^2}} + \frac{V_{max}}{a}$$

$$\text{Time delay} = \sqrt{-\frac{2}{a} (S+1) + \frac{2D}{a} + \frac{V_{max}^2}{a^2}} -$$

$$\sqrt{\frac{2}{a} (S) + \frac{2D}{a} + \frac{V_{max}^2}{a^2}}$$